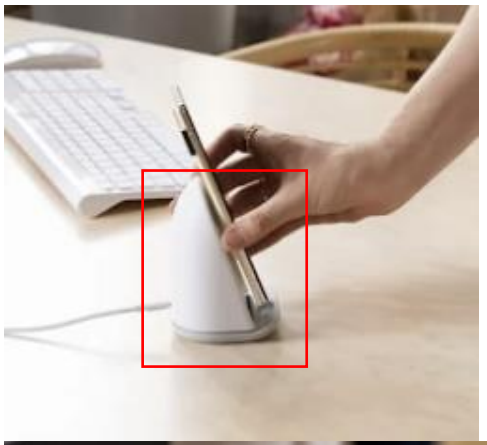
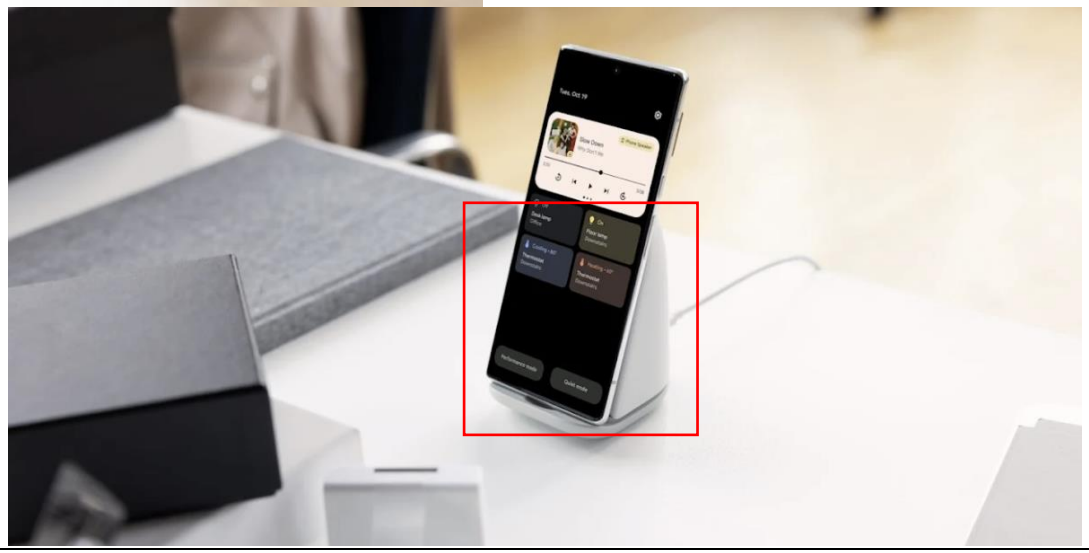
US10600547B2	Google Pixel Stand 2nd Gen
1. A coil module for an induction type power supply system comprising:	*A coil module (e.g., Google Pixel Stand 2nd Gen) for an induction type power supply (e.g., Wireless power transfer).* *The Google pixel stand 2nd Gen is Qi Certified that supports magnetic induction.*  

Charging

Up to 23W for compatible Pixel phones.

Supports up to 15W (EPP) for Qi Certified devices.[5]

Source: https://store.google.com/us/product/pixel_stand_2nd_gen?hl=en-US



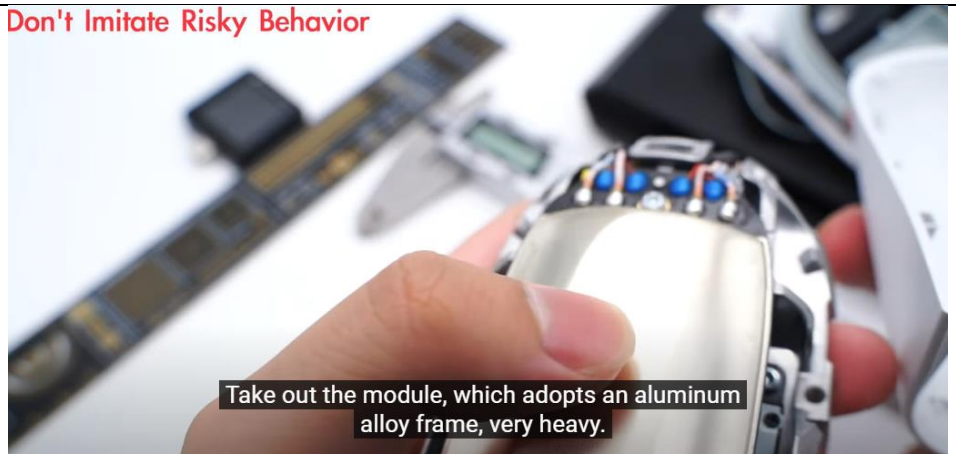
MAGNETIC INDUCTION

Sometimes it looks like magic but it's really just basic science.

See how Qi works →

Source: https://www.wirelesspowerconsortium.com/standards/qi-wireless-charging/

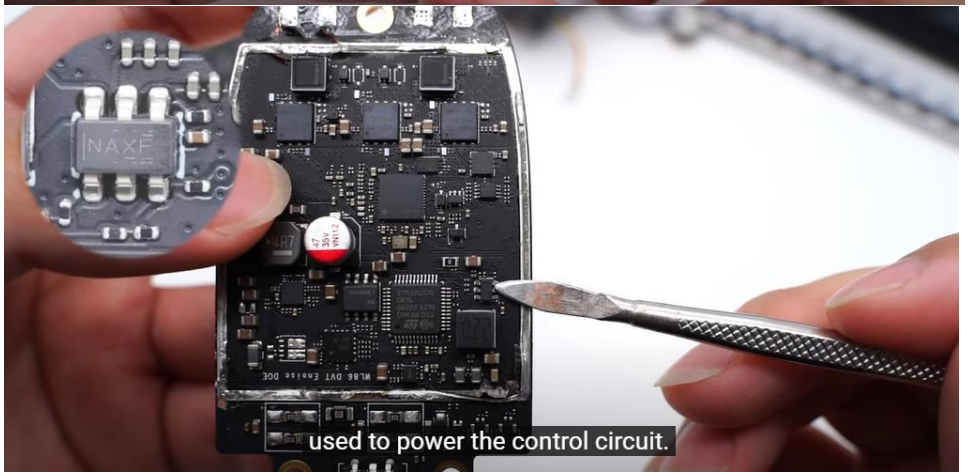
	How Qi Works Magnetic induction is a technology that you will probably remember from your physics classes at high school. You need two coils, a transmitter coil and a receiver coil. An alternating current in the transmitter coil generates a magnetic field which induces a voltage in the receiver coil. This voltage can be used to power a mobile device or charge a battery. Source: https://www.wirelesspowerconsortium.com/knowledge-base/magnetic-induction/how-qi-works/
a metal frame;	*Google pixel stand 2nd Gen has an aluminium frame (e.g., metal frame).* 

	 Source: https://www.youtube.com/watch?v=WcAi8wGLNqg (3:14 Mins – 3:20 Mins)
a coil, disposed on a first surface of the metal frame;	*A coil (e.g., charging coil), disposed on a first surface of the metal frame (e.g., one side of the aluminium alloy frame).* 

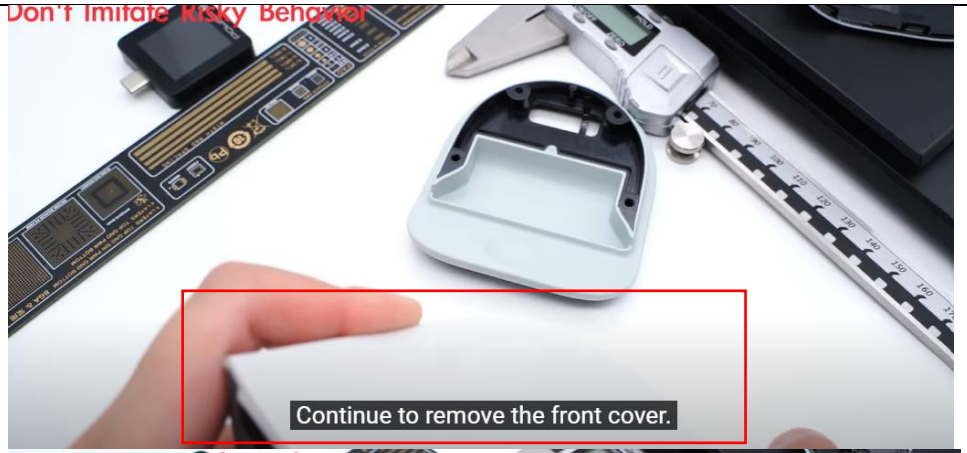
	 Source: https://www.youtube.com/watch?v=WcAi8wGLNqg (2:27 Mins – 3:07Mins)
a circuit board, disposed on a second surface of the metal frame, the circuit board comprising a control circuit for controlling operations of the coil; and	*The Google Stand 2nd Gen includes a circuit board (e.g., PCB), disposed on a second surface (e.g., second side) of the metal frame (e.g., Aluminium frame), the circuit board comprising a control circuit for controlling operations of the coil (e.g., the circuit is connected with the coil therefore, it is inferred that the control circuit controls the operations of the coil); and* 


Don't Imitate Risky Behavior
Let's separate the PCB.


Don't Imitate Risky Behavior

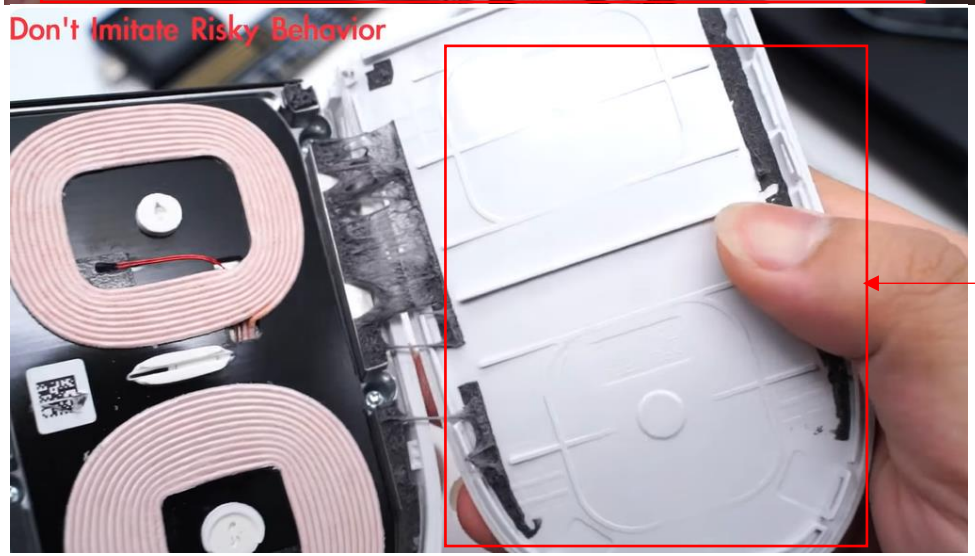
	  Source: https://www.youtube.com/watch?v=WcAi8wGLNqg (3:22 Mins – 7:40 Mins)
an upper lid, for covering the coil, the upper lid composed of a non-metal material and	*The Google Pixel 2nd Gen has an upper lid (e.g., front cover), for covering the coil (e.g., wireless charging coil), the upper lid composed of a non-metal material (e.g., plastic) and having an arc structure.*

having an arc structure.	 	


Don't Imitate Risky Behavior
Continue to remove the front cover.

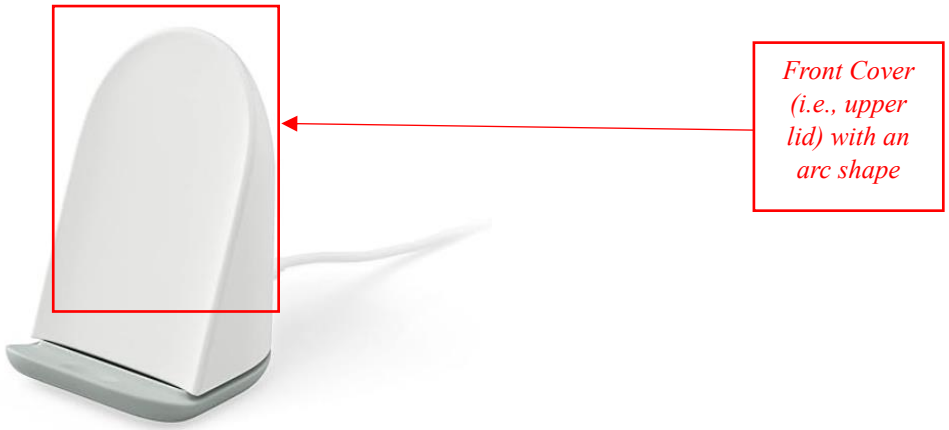

Don't Imitate Risky Behavior





Front Cover (i.e., upper lid) with an arc shape

Source: https://www.youtube.com/watch?v=WcAi8wGLNqg (1:30 Mins -2:38 Mins)



Source: https://www.amazon.com/Google-Pixel-Stand-2nd-Gen/dp/B09HL8RZ6W?keywords=google+pixel+stand&qid=1647566213&sbo=RZvfv//HxDF%2BO5021pAnSA%3D%3D&sr=8-3&linkCode=sl1&tag=chargerlab0d-20&linkId=44ad6125bb858da2fb447dc5086cea95&ref_=as_li_ss_tl